

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I Bhuvankumar J(192524040) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO2: Assessment Tool 1 - Short Answer1. Cloud computing

cloud computing is like a delivery service which delivers Server's, Database's and etc. It provides Networking space which can be bought and does not need for a physical Storing device in your PC. ~~Yes Space is~~ Storing space can be bought and saved online where the cloud computing or cloud Storage provider will store the data in data centers. Cloud computing ~~can~~ is a pay what you use service. We only need to pay for what amount of Storage we use. It is scalable and highly available.

characteristics :

- * Cloud computing is more efficient because we only pay for what amount of Storage space we use. It can be as the space can be easily increased or decrease based on the demand.

- * uploading and retrieving data is more efficient now, this saves more time.

- * Data sharing is also very efficient, as the networking is done online. many systems can be hosted with single server.

- * Rapid elasticity based on customer's Demand.

2.

The evolution of cloud computing from hardware evolution to internet software and the changes are listed below.

i) Main frame Era:

In 1960, A Big main Screen will be there where everyone will work with one big powerful screen, these kind of screens are used by business persons in those time. This system will be big and cost high and also requires maintenance.

ii) Personal computer era (1980's)

The time personal computers came. Every person willing to have a system can own one with storage hardware's.

iii) Networking and internet:

Introduction of internet made data sharing more easy and efficient. without any hardware we can share data and also obtain data from websites.

iv) virtualization:

one server can be divided into multiple virtual machine system. This is the most efficient way because each have an independent system and OS. but storage data's are stored in single server.

4) cloud computing:

This is the modern way of storing data.

Data Storage can be bought online and the amount you need and can store data without you having a hardware for storing data.

4. Data center's :

Data center is a place where many servers, database's and network management device's are kept. The provider of cloud will provide storage space from here.

- * The ^{Space} ~~Data~~ can be given more or less by customer demand.
- * They have high speed internet, backup servers, disaster management, Backup electricity to provide the server without lag and ~~to~~ not to cause server down issue.
- * They have high encryption, so data can't be hacked or ~~lost~~ lost. They provide ~~safe~~ encryption by firewalls.
- * There are ~~cost~~ coolants to cool the server so there won't be any issue because of heat.

3.

Server Virtualization:

Server virtualization is the process of dividing one physical server into multiple virtual machines, where each VM works like an independent computer with its own OS and application.

It improves hardware utilization by allowing multiple workloads to run on a single server, reducing the need for additional physical hardware.

Virtualization helps reduce cost, saves energy, simplifies server management and supports faster development of application.

Virtualization is enabling tech that creates virtual resources such as servers, storage and networking. It forms the foundation of cloud computing.

Difference:

Virtualization focuses on creating and managing virtual resources while cloud computing focuses on delivering those resources as services (IaaS, PaaS, SaaS) to users over the internet.



5. All these (IaaS, PaaS, SaaS and XaaS) are cloud computing services.

IaaS:

Infrastructure as a Service

- * It provides Hardware Storage which will be managed by the provider
- * The application's and web application can be managed by the user.

SaaS:

Software as a service:

- * This service provides and manages the software. using this service we can buy and use storage space without hardware.

eg: Gmail

PaaS

Platform as a Service:

This service not only handle's and have softwares, websites & apps.

eg: Google one.

XaaS

Everything as a Service:

Anything you use is a Service here like for buying storing any process left by rest of the Service's will come under XaaS.

eg:

→ All these Service combined together gives us cloud computing service.